EF

CHEM P 2 SET 1 2 3 4 CONFIDENTIAL

COMMON Apparatus and materials

- Access to a water supply and a source of heat (Bunsen burner)
- Test-tube rack, clean test tubes, boiling tubes, red and blue litmus papers, glass rod
- Burette (0–50 cm^3), pipette (20 cm^3 or 25 cm^3), two or three 250 cm^3 conical flasks
- Titration tile/WHITE PAPPER
- Phenolphthalein indicator

SET ONE

Item	Supply per candidate
Item 1: solid P_1	About 2.0 g of copper(II) sulfate pentahydrate, $CuSO_4 \cdot 5H_2O$, in a clean dry specimen bottle labelled P_1
Item 2: solid Z_1	Exactly 1.16 g of pure magnesium hydroxide, $Mg(OH)_2$ powder, in a stoppered weighing bottle labelled Z_1
Item 2: FA1	HCl, approx. 0.20 mol dm^{-3} (7.3 g dm^{-3}) Dilute 20.0 cm^3 of concentrated hydrochloric acid to 1 dm^3 of solution
Item 2: FA2	NaOH, 0.10 mol dm^{-3} (4.0 g dm^{-3}) Dissolve 4.0 g of NaOH and make up to 1 dm^3 of solution

Substitution: if copper(II) sulfate pentahydrate is unavailable, copper(II) chloride dihydrate, $CuCl_2 \cdot 2H_2O$, may be used. It readily gives the characteristic tests for Cu^{2+} and Cl^- ions.

SET TWO

Item	Supply per candidate
Item 1: solid P_2	About 2.0 g of zinc sulfate heptahydrate, $ZnSO_4 \cdot 7H_2O$, in a clean dry specimen bottle labelled P_2
Item 2: solid Z_2	Exactly 1.17 g of pure aluminium hydroxide, $Al(OH)_3$ powder, in a stoppered weighing bottle labelled Z_2
Item 2: FA1	HCl, approx. 0.20 mol dm^{-3} (7.3 g dm^{-3}) Dilute 20.0 cm^3 of concentrated hydrochloric acid to 1 dm^3 of solution
Item 2: FA2	NaOH, 0.10 mol dm^{-3} (4.0 g dm^{-3})

Item	Supply per candidate
	Dissolve 4.0 g of NaOH and make up to 1 dm^3 of solution

Substitution: if zinc sulfate heptahydrate is unavailable, zinc chloride, $ZnCl_2$, or zinc nitrate hexahydrate, $Zn(NO_3)_2 \cdot 6H_2O$, may be used. Both give the characteristic tests for Zn^{2+}.

SET THREE

Item	Supply per candidate
Item 1: solid P_3	About 2.0 g of iron(II) ammonium sulfate hexahydrate, $FeSO_4 \cdot (NH_4)_2SO_4 \cdot 6H_2O$, in a clean dry specimen bottle labelled P_3
Item 2: solid Z_3	Exactly 1.25 g of pure calcium carbonate, $CaCO_3$ powder, in a stoppered weighing bottle labelled Z_3
Item 2: FA1	HCl, approx. 0.20 mol dm^{-3} (7.3 g dm^{-3}) Dilute 20.0 cm^3 of concentrated hydrochloric acid to 1 dm^3 of solution
Item 2: FA2	NaOH, 0.10 mol dm^{-3} (4.0 g dm^{-3}) Dissolve 4.0 g of NaOH and make up to 1 dm^3 of solution

Substitution: if iron(II) ammonium sulfate hexahydrate is unavailable, other ammonium iron(II) sulfate crystals may be used. They readily give tests for Fe^{2+}, NH_4^+ and SO_4^{2-}.

SET FOUR

Item	Supply per candidate
Item 1: solid P_4	About 2.0 g of ammonium sulfate, $(NH_4)_2SO_4$, in a clean dry specimen bottle labelled P_4
Item 2: solid Z_4	Exactly 1.15 g of pure magnesium carbonate, $MgCO_3$ powder, in a stoppered weighing bottle labelled Z_4
Item 2: FA1	HCl, approx. 0.20 mol dm^{-3} (7.3 g dm^{-3}) Dilute 20.0 cm^3 of concentrated hydrochloric acid to 1 dm^3 of solution
Item 2: FA2	NaOH, 0.10 mol dm^{-3} (4.0 g dm^{-3}) Dissolve 4.0 g of NaOH and make up to 1 dm^3 of solution

Substitution: if ammonium sulfate is unavailable, ammonium chloride, NH_4Cl, may be used. It gives tests for NH_4^+ and Cl^- ions.